Math 3
VII-1 Solving Linear Equations: MultiStep Equations

Name:_____
Date: _____ Per: ___

Imagine a balanced scale with 5 bags and 2 cubes on one side, and 3 bags and 4 cubes on the other. Model this:



Describe the steps you would take to determine how many cubes each bag contains.

Now imagine another scale with 12 bags and 5 bags on one side, with 6 bags and 17 cubes on the other. You probably don't want to model this. Why not? What should be done instead?

Equations with Variables on Both Sides

Step 1: Simplify what you can on each side of the equation

Step 2: Move all terms with variable to one side

Step 3: Move all constant terms to opposite side of variables

Step 4: Solve

Examples:

1) $10 - 4x = -9x$

2) $-2x = 3x + 10$

3) $10x - 8 + 14x = 40$

4) $\frac{1}{2}(6x - 4) = -5x + 1$

5) $-\frac{3}{4}(8x + 12) - 3(x - 3) = 0$

Turn Over

You Try/Practice:

For problems 1 - 12, solve for x.

1) $5x + 3 - 3x = -7$

2) $12x + 14 + 10x = 80$

3) $11x - 9 - 7x = 15$

4) $3 + 4(x + 5) = 31$

5) $27 = 3 - 3(6 - 2x)$

6) $7x - 12 = 6 + 2x$

7) $3(4x + 2) - 2(5x - 3) = 12$

8) $-2(4x - 5) = 8 - 6(2x + 1)$

9) $3(x + 12) = 8 - 4x$

10) $6(x - 10) = 5x - 42 - 2x$

11) $2(4x + 2) = 4x - 12(x - 1)$

12) $\frac{2}{3}(3x + 9) = -2(2x + 6)$